Rate-Distortion Loss Attribution for AV1 Intra Partition Decisions

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***Abstract*—The AOMedia Video 1 (AV1) codec decides how to split each intra-frame super block through a recursive search that evaluates ten partition shapes per node, and learned pruners are the dominant strategy to shorten that search. Every such pruner couples the quality of the source that scores a node with the aggressiveness of the policy acting on that score, so the reported time savings do not identify which information carries the decision. This paper presents the Representation Probe for Partitioning (RPP), an offline attribution protocol for AV1 intra partitioning that reads the rate-distortion loss of competing representations at matched search-cost reduction. RPP fixes one pruning policy, sweeps its confidence threshold, and charges each forced decision the overhead recorded by an exhaustive reference search, so that representations are compared at the same amount of work skipped. Experimental results over 3.81 million held-out decision nodes show that twenty-four compact descriptors reduce the loss of isolated block variance by 7.0 times, that eight causal partitioning columns reduce it by a further 1.60 times, and that four quantization and position columns add nothing. A ConvNeXt with 28.1 million parameters reading raw pixels stays 4.1 times behind the compact descriptors, which hold 2481 times fewer parameters, and doubling its fusion width makes it worse. The protocol is ordinal and does not convert into Bjontegaard delta bitrate. To the best of the authors' knowledge, this is the first work to attribute rate-distortion loss to information access in AV1 partitioning under matched cost.**

***Index Terms*—Video coding, AV1, intra-frame partitioning, feature representation, model capacity, machine learning.**

I. Introduction

Video traffic keeps its position as the dominant share of Internet usage, and the proportion of users consuming online video every month has remained above ninety percent since 2022 [1]. Two families of codecs answer that demand: the standardization path produced High Efficiency Video Coding [2] and Versatile Video Coding [3], both developed jointly by ISO/IEC MPEG and ITU-T VCEG, while the Alliance for Open Media released the AOMedia Video 1 (AV1) in 2018 as an open and royalty-free codec [4], drawing on VP9, Thor, and Daala.

AV1 follows the hybrid block-based coding flow, combining intra-frame and inter-frame prediction, transforms, quantization, in-loop filters, and entropy coding, and it introduced new tools at every one of those stages [5], [6]. Among them, the partition structure was extended to ten shapes per node, against the four of VP9. That coding efficiency is paid for in computational effort, and comparative studies report AV1 encoding times far above those of its competitors [7], [8], with the gap widening at the highest quality configuration.

Inside intra-frame coding, the recursive partition search is the single most expensive stage at that configuration. The search visits a quad-tree of square nodes, evaluates up to ten shapes at each one through full rate-distortion optimization [9], and descends into every node it does not resolve, so the cost grows with the number of nodes rather than with the difficulty of the content.

Several works attack this cost with learned decisions. In AV1, tool selection guided by user priority and by coding efficiency was proposed in [10], [11], a fast intra mode decision based on the accuracy of a rate-distortion model in [12], and decision trees for inter prediction in [13]. The same problem in Versatile Video Coding was addressed with light gradient boosting [14], random forests [15], and variance and gradient heuristics [16], while dedicated hardware architectures targeted the AV1 intra prediction modes [17], [18]. These works can be classified as coding-efficiency oriented [11], [12], [14]–[16], hardware oriented [17], [18], or aimed at a different coding stage [10], [13]. All of them report time savings obtained under a policy of their own choosing, and none of these works separates the quality of the scoring source from the aggressiveness of the policy, nor measures which information is decisive for the partition choice.

This paper presents the Representation Probe for Partitioning (RPP), an offline attribution protocol for AV1 intra partitioning. The main strategy of this solution is to hold the pruning policy and the amount of skipped search work fixed, and to vary only what a representation is allowed to see, so that the resulting difference in rate-distortion loss is attributable to information access alone. Under this protocol, compact descriptors outperform a high-capacity convolutional model reading raw pixels, and the causal partitioning neighborhood carries decision information that luminance does not provide.

II. Intra Partition Search in AV1

AV1 encodes each frame as a grid of super blocks of 64×64 samples, and each super block is recursively divided by a search over a quad-tree of square nodes of 64, 32, 16, and 8 samples. At every node the encoder may evaluate ten partition shapes, organized in four groups: the undivided block, the quaternary split, the two rectangular shapes, and the six extended shapes.

The evaluation order inside a node is fixed. The undivided candidate is evaluated first and produces a complete rate-distortion cost; the quaternary split follows, and opening it triggers the recursion into the four children; the rectangular shapes come next, and the extended shapes last.

This order defines when each kind of information becomes available, which is the property this work exploits. Before the search of a node begins, the encoder already holds the source luminance of the block, the frame quantization index, the position of the node, and the partition decisions of the causal neighbors above and to the left, but it does not yet hold any rate-distortion cost for the current node. Fig. 1 summarizes this timeline.

The action studied in this paper is the earliest one available: committing the node to the undivided shape before any rate-distortion evaluation is paid, and therefore skipping every remaining candidate of the node and the whole subtree below it. This single action has two measurable consequences, the search work that is no longer performed and the rate-distortion cost that is lost whenever the discarded subtree was the optimal choice.

This work is focused on that commitment at the pre-search insertion point of the intra-frame partition tree. Inter-frame prediction, transform type selection, quantization, in-loop filters, and entropy coding are not included in the loop attacked here, and therefore they are not focused on this work. Blocks of 8×8 samples are terminal leaves whose reference label is invariably the undivided shape, and they are excluded from modeling while kept in the cost accounting.

III. Proposed RPP Protocol

RPP is a measurement protocol, not an encoder solution. It compares representations by what they are allowed to see, under one fixed policy and at a matched amount of skipped search work.

A. Representations Declared by Information Access

The representations are declared by information access rather than by architecture. The first is the isolated variance of the block, the classical descriptor of partition heuristics. The second, denoted A, is a twenty-four column vector of compact

PLACEHOLDER — information-availability timeline inside one node: causal context and source luminance available before the search; undivided cost available after the first candidate; subtree costs available only after the recursion. The pre-search instant is marked.

Fig. 1. Information available to a pruner at each instant of the search of a partition node, and the pre-search instant addressed in this work.

luminance descriptors of the block and of its hierarchical context: variance and its per-quadrant dispersion, gradient energy and orientation, row and column profiles, edge statistics, direct-current level, and the contrast against the parent and sibling blocks. Three of those columns do not derive from luminance, namely the normalized quantization index and the two coordinates of the node inside the 64-sample unit.

Block B adds eight columns of causal partitioning neighborhood: availability of the above and left neighbors, their already decided widths and heights in logarithmic scale, the relative granularity of the neighborhood, and its anisotropy. Block C adds four columns of effective quantization and position, namely the dequantization step of the direct-current coefficient, the normalized position within the frame, and the depth of the node. Four rungs are built from these blocks, namely A, A+B, A+C, and A+B+C, the last one holding thirty-six columns. All of these columns are natively available during encoding at the pre-search instant, ensuring zero computational overhead for feature extraction.

The remaining representation is the deep pixel arm, a multi-level convolutional model of the ConvNeXt family [19] that mirrors the partition hierarchy. Its input is a two-channel tensor of 64×64 samples, formed by the super-block luminance and a constant quantization plane. Three stages of the trunk are merged top-down, so that each cell of the resulting map holds both its immediate neighborhood and the context of the whole super block, and per-level heads read average poolings of it. This arm is present as a test of the hypothesis that raw pixels plus representation capacity suffice, and not as an architectural baseline. A random scorer is included as a control.

The four rungs are multilayer perceptrons with two hidden layers of 64 and 32 rectified units, one network per block size, trained with hard-label cross entropy and AdamW [20] at a learning rate of 1×10^{-3}, batches of 4096 samples, and a fixed budget of 30 epochs. Every rung is trained with this one recipe and differs only in which columns it reads, which is what makes the verdict fall on information rather than on capacity or on training procedure. Each rung is trained with three seeds and the reported value is their mean.

B. Policy, Cost Model, and Loss

The policy is fixed for every representation. A node n is committed to the undivided shape whenever the score assigned to that class exceeds a confidence threshold τ, in which case the node contributes only the cost of its undivided candidate and the recursion stops; otherwise the node is evaluated in full and the search descends. The rate-distortion loss charged to a committed node is the overhead of that commitment against the optimal subtree, as recorded by the exhaustive reference search,

$$r(n) = J_{\mathrm{none}}(n) - J^{*}(n), \tag{1}$$

where $J_{\mathrm{none}}(n)$ is the rate-distortion cost of the undivided candidate and $J^{*}(n)$ is the cost of the optimal subtree rooted at n. By construction $r(n) \geq 0$, and it is zero whenever the undivided shape already was the reference decision. No encoding is repeated, since the whole protocol is a counterfactual replay over the log of one exhaustive search.

The search work of a node of dimension n is counted analytically as

$$c(n) = k_n\, n^2, \quad k_n = 9 \text{ for } n \in \{64, 32, 16\},\ k_8 = 4, \tag{2}$$

that is, the number of shape candidates a full search would evaluate, weighted by the area of the block. The cost reduction $\Delta(\tau)$ is the percentage difference between the work performed by the policy and the work of the full search. It is a counted quantity and not wall-clock time, and it is not an estimate of wall-clock time either.

The rate-distortion loss of a representation at a threshold τ is

$$L(\tau) = 100\, \frac{\sum_{n \in P(\tau)} r(n)}{\sum_{n \in N} J_{\mathrm{none}}(n)}, \tag{3}$$

where $P(\tau)$ is the set of committed nodes and N is the set of all decision nodes. The denominator is accumulated before any threshold is fixed and stays constant along the sweep, which is the condition for L to be readable as a frontier, since it tends to zero as pruning tends to zero.

Two properties of L restrict its reading, and they are stated before any number is reported. First, the denominator aggregates the undivided costs of the three hierarchy levels, which cover the same image area, so it is systematically larger than the cost of coding the frame and L is systematically smaller than any efficiency loss observable in the encoder. Second, the replay discounts recorded values, whereas pruning inside the encoder also alters the reconstruction that serves as reference to later blocks, a propagation absent here. Consequently, no conversion factor between L and the Bjøntegaard delta bitrate [26] is postulated, and L is used strictly in an ordinal regime, in which every ratio is taken between representations read at the same operating point. Each representation therefore yields a curve rather than a point, and the reference reading in this paper is $\Delta = 25\%$.

C. Dataset and Experimental Setup

The reference labels were extracted from the libaom v3.10.0 reference software [22], instrumented under a compile-time guard so that the production binary remains byte-identical to the original. The encoder was configured for All-Intra following the AOM Common Test Conditions [23] at `cpu-used=0`, which enables the full rate-distortion search and therefore makes the recorded decisions optimal by construction.

The corpus comprises sixteen UVG sequences [24] at 4K resolution (3840×2160), 8 bits and 4:2:0 chroma subsampling, obtained from the XIPH repository [25]. Each sequence was encoded at four quantization points (`cq-level` 20, 32, 43, and 55) over five frames sampled uniformly along the whole clip, since consecutive frames are nearly identical under All-Intra. The resulting dataset holds 26.98 million samples, of which 10.07 million are decision nodes. The split is made by sequence and was frozen before any measurement, with ten sequences for training, three for validation, and three for the reserved test set. No video sequence was reused across the training, testing, or evaluation datasets, thus avoiding data leakage and overfitting. All results reported here are measured on the six held-out sequences, which contribute 226 447 super blocks and 3 808 703 decision nodes. The models were implemented in PyTorch [21].

D. Fair Treatment of the Deep Arm

Since the deep arm carries the hypothesis that raw pixels plus capacity suffice, a negative result on it is only meaningful if the arm was treated fairly, and three measures were taken. First, it was retrained on the same corpus and under the same split as the rungs, after an audit of an earlier checkpoint showed that it had been trained on a smaller corpus with a single quantization point and with two of the six held-out sequences contaminated. Second, capacity was controlled by doubling the fusion width from 128 to 256 channels, and the wider model is 1.0% worse in validation loss, which establishes that capacity is not the binding constraint.

Third, the training objective was ablated. A cost-sensitive cross entropy weighting each node by $1 + \alpha\, \hat{r}(n)$, with $\hat{r}$ the normalized loss of (1), was compared against plain cross entropy, obtained by setting $\alpha = 0$. Under identical corpus, schedule, and selection criterion, the cost-sensitive objective at $\alpha = 3$ is better at every operating point from 10% of cost reduction onward, and it is the deep result discussed in Section IV. This inverts a conclusion previously drawn by the authors from an uncontrolled comparison, so the value reported here is the most favorable one measured for the deep arm.

IV. Results and Discussion

The representations were evaluated with the protocol of Section III over the six held-out sequences and 3 808 703 decision nodes, under the All-Intra configuration of libaom v3.10.0 at `cpu-used=0`. Table I reports the loss L of (3) at five matched cost reductions, in units of 10^{-3} percent of the aggregated undivided cost, where lower is better, and Fig. 2 shows the corresponding frontiers.

The main conclusion when observing Table I is that the ordering of the representations is preserved across the whole

TABLE I
RATE-DISTORTION LOSS AT MATCHED COST REDUCTION, IN UNITS OF 10^{-3} PERCENT OF THE AGGREGATED UNDIVIDED COST

Representation	**10%**	**15%**	**20%**	**25%**	**30%**
Random control	196.3	298.9	406.6	516.6	634.2
Variance	4.3	12.7	25.0	37.4	55.5
ConvNeXt, plain CE	6.5	10.5	16.4	27.2	44.2
ConvNeXt, width 256	7.5	12.2	19.4	29.4	44.4
ConvNeXt, cost-sensitive	5.4	8.0	14.0	21.9	37.9
A (24 columns)	0.5	1.3	2.8	5.3	9.1
A+C (28 columns)	0.5	1.2	2.7	6.4	12.2
A+B+C (36 columns)	0.2	0.6	1.8	4.0	7.8
A+B (32 columns)	**0.2**	**0.5**	**1.5**	**3.3**	**6.9**

PLACEHOLDER — rate-distortion loss versus matched cost reduction, logarithmic vertical axis, one curve per representation, with seed-amplitude bars on the four rungs.

Fig. 2. Rate-distortion loss as a function of matched search-cost reduction. Bars on the rungs show the amplitude over three training seeds.

range, so it is not an artifact of one operating point. At 25%, isolated variance loses 37.4 units while the twenty-four compact columns of rung A lose 5.3, a reduction of 7.0 times, against the 516.6 units of the random control. This is the largest single gain in the table, and it contradicts any reading in which the information available in the pixel domain would be exhausted by variance.

Adding the eight causal partitioning columns to rung A lowers the loss from 5.3 to 3.3 units, a further reduction of 1.60 times. The separation is robust to initialization, since the worst of the three seeds of A+B, at 3.6, is still better than the best of the three seeds of A, at 4.9. This is the central result of this work, because those eight columns describe the partition choices of the already coded neighbors and are not computable from the luminance of the block under any transformation. The information they carry is therefore produced by the encoding process itself.

Conversely, the four columns of effective quantization, frame position, and node depth add nothing. Rung A+C reaches 6.4 units against 5.3 for rung A at 25%, and its seed amplitude is 4.5 units against 0.8 for rung A, so the two distributions overlap and the statement supported by the data is that block C does not add information while destabilizing the fit. Once combined with the causal columns the effect is no longer ambiguous, as A+B reaches 3.3 units against 4.0 for A+B+C, all nine pairwise seed comparisons agree, and Fig. 2 shows A+B dominating A+B+C across the entire frontier. The thirty-six column vector is therefore not the offline optimum, although this superiority of A+B has not been verified inside the encoder.

The deep arm does not support the hypothesis that raw pixels plus representation capacity suffice. In its most favorable configuration it reaches 21.9 units at 25%, which is 4.1 times worse than rung A, although it holds 28 128 638 parameters against the 11 337 of the three networks of rung A, a factor of 2481. Doubling the fusion width degrades the result rather than improving it, at 29.4 against 27.2 units under the same objective. This may be attributed to the information that is missing rather than to the capacity that is available, since the deep arm reads the same pixels as rung A and the gap is not closed by capacity, whereas eight columns of encoder state close a further factor of 1.60 with 4291 parameters per level. It is important to emphasize that the deep arm is therefore not an upper bound of the pixel domain, since a model outperformed by another with strictly smaller information access establishes no performance ceiling. The only genuine upper bound here is the reference search itself, whose loss is zero by construction.

Three limitations qualify these results and are reported as part of them. The protocol is ordinal, as established in Section III-B, so none of these values converts into a Bjøntegaard delta bitrate or predicts what an encoder would deliver. The three seeds provide an amplitude and not a confidence interval. The objective ablation of the deep arm covers two points, $\alpha = 0$ and $\alpha = 3$, and the curve in α was not swept. The comparison is nevertheless conservative in the direction that matters, since every one of these choices was resolved in favor of the deep arm.

V. CONCLUSIONS

This paper presented RPP, an offline attribution protocol that measures the rate-distortion loss of competing representations of the AV1 intra partition decision at matched search-cost reduction, reaching the conclusion that the causal state of the encoding process carries decision information that luminance alone does not provide. The protocol holds the pruning policy and the training recipe fixed and varies only the information a representation may read, charging every forced decision the overhead recorded by an exhaustive reference search. It was evaluated over 3.81 million held-out decision nodes of sixteen UVG 4K sequences encoded with libaom v3.10.0. Twenty-four compact descriptors improve on isolated variance by 7.0 times, eight causal partitioning columns improve on those by a further 1.60 times, and four quantization and position columns add nothing. A ConvNeXt with 28.1 million parameters reading raw pixels remains 4.1 times behind the compact descriptors, and doubling its width makes it worse, so capacity is not the binding constraint. Besides, to the best of the authors' knowledge, this is the first work to attribute rate-distortion loss to information access in AV1 partitioning.

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COMPLIANCE WITH ETHICAL STANDARDS

This work is a computational study over publicly available video sequences and does not involve human participants or animals. The authors declare that they have no conflict of interest.

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